Pilot Project Proposal

Climate Action for Human Security in Iraq's Al Hawizeh Marshlands

Iraq pilot grant under regional SDG Climate Facility project: Climate Action for Human Security[1]

- $922,000 grant from SDG Climate Facility regional project
- UNDP – Iraq CO Contribution: $150,000
- Timeframe for implementation: 36 months (3 Years)
- Implementing partner: UNDP Iraq
- Required focus on cross-cutting 'climate security'

I. Interlinked challenges – Background and Context

1. The Al Hawizeh Marshes are located in the Missan governorate in southeast Iraq just east of the Tigris River. Together with the adjacent Central and Al Hammar marshes within the Thi Qar and Al Basrah governorates, respectively, they are at the southern tip of large alluvial plain fed by the flows of the Tigris and Euphrates rivers and their distributaries (see Figure 1). The three marsh areas consist of a set of interconnected, shallow, freshwater lakes, marshes and seasonally inundated large, flat floodplains.

2. Today, the Al Hawizeh marshes suffer from several interlinked development challenges. Foremost of these is addressing the immediate livelihood needs of IDPs that have returned to the marshes from other governorates. Returnees have faced the problem of severe environmental degradation and a near total absence of the infrastructure required to support everyday life—lack of safe drinking water, power supply, refuse collection and sewage treatment facilities, schools, clinics, and other public facilities (SCECI, 2006). At present, these conditions have experienced some improvements

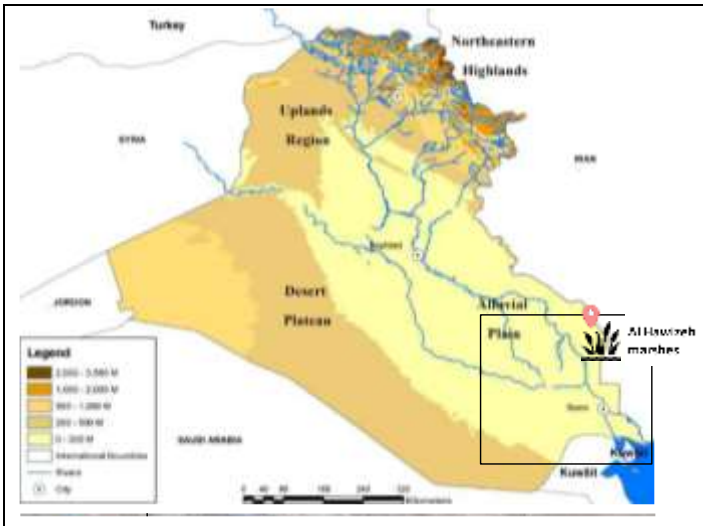


Figure 1: Top: Major physical regions of Iraq (RoI, 2016)

[1] The 'SDG Climate Facility project: Climate Action for Human Security' (Sida: $8.8 million; UNDP: $600,000) is a multi-stakeholder partnership comprising League of Arab States, Arab Water Council, UNDP, UNDRR, UNEP-FI, UN-Habitat and WFP. The regional project includes regional Arab-wide as well as national-level activities.

owing to a range of marsh restoration initiatives. Nevertheless, most marsh villages do not have access to clean water supply or wastewater treatment or disposal. Bath water is returned to the marsh and village houses lack toilets, resulting in sewage input directly into the wetlands (USAID, 2004). A basic summary of infrastructure availability is provided in Box 1.

3. Second, social fragility remains high throughout Iraq (FFP, 2019); with its inverse – social cohesion - remaining low. In the marshlands - as well as the country at large – communities are facing underlying and often overlapping political, socio-economic and security challenges that can adversely impact efforts to promote sustainable development. Advancing connectedness and solidarity among groups and villages in marshland society remains challenging. Apart from deliberate interventions and processes to reduce socioeconomic disparities, the risks are high that social and environmental shocks could diminish prospects for improved social cohesion and good governance. In a worst case, there is a risk for a break-out of conflict and violence at the local level as the lack of social coherence, widespread poverty, and lack of near-term livelihood improvement could contribute to recruitment activities of illegal armed groups. This holds especially true for young people.

4. Third, climate change poses a mounting threat that exacerbates the above challenges. Recent climate trends averaged over the 1980-2017 period from meteorological stations in close proximity to the marshes (i.e., Chibayesh in the Central marshes and Suq Al Shoyokh in the western Hammar marshes) show that average maximum temperatures have increased in the marshes at the rate of about 0.27°C per decade, while monthly precipitation has decreased at the rate of about 0.66 mm/month per decade. Concurrently, the livelihoods of Al Hawizeh marsh dwellers are being subjected to increasing climate-related impacts, as outlined below.

5. The aforementioned issues are context-specific to Iraq and the Marshlands and show how several Sustainable Development Goals (SDGs), the Paris Agreement and Sendai Framework on Disaster Risk Reduction, are interlinked in a way that requires cross-cutting climate-security solution approaches. The latter needs to enhance social cohesion and stability towards SDG-16, enhance resilience and strengthen local climate action towards SDG-13, and improve the provision of clean drinking water and

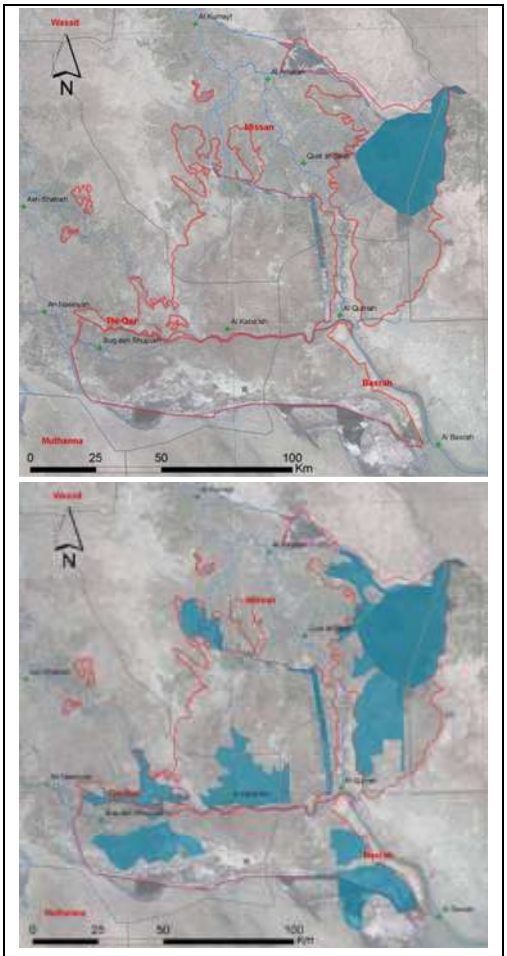


Figure 2: Top: Pre-1973 extent of Iraqi marshlands demarcated by red lines; remaining marshland in 2003 illustrated by blue-filled area (adapted from RoI, 2006); Bottom: Marsh extent in 2005 relative to the 1973 spatial extent (RoI, 2014)

Box 1: Infrastructure overview, Al Hawizeh marshes (Thi Qar University & UNEP, 2007)

- *Water supply:* There is acute drinking water supply scarcity in most villages. The Euphrates and Tigris are too saline, turbid, and polluted. Reverse osmosis units, water pipelines from other areas, tanker water, and groundwater make up 68% of drinking water with the balance made up of river/marsh water.
- *Sanitation:* Waste management conditions are de minimis in most Al Hawizeh marsh villages. About 65% of the population disposes of sewage just outside their houses or by digging small pits. Latrines and septic systems service about 34% and 4% of the villages, respectively.
- *Primary schools:* A large proportion of villages in Missan, about 41%, do not have local primary schools. For those villages with primary schools, the student-teacher ratio tends to be high, about 18:1.
- *Electricity:* There are no public generators in the region and many 29% of Missan villages do not have access to any electricity. Those villages with central grid connections suffer from chronic blackouts.

sanitation towards SDG-6. In addition, SDG-5 on gender equality and SDG-1 on poverty go hand-in-hand with climate security and resilience in the Marshlands, as elucidated further below.

6. The Al-Hawizeh marsh is hydrologically distinct from the other marshes as it straddles the Iraq-Iran border and in addition to being fed by distributaries of the Tigris river, it is also fed by the Karkheh River in Iran, making it a transboundary marshland. Prior to the 1980s, the three marshes were fed by approximately 55.1 billion cubic meters (BCM) per year from the Tigris-Euphrates river basin (UNESCWA and BGR, 2013a, 2013b) plus some additional amounts from Karkheh river basin (Marjanizadeh et al., 2009). Beginning in the 1980s and 1990s, however, large dams in the upper reaches of the Euphrates and Tigris started to change the water distribution throughout the Tigris-Euphrates river basin, strongly impacting downstream water availability and use (Hashim, *et al*., 2019; Shamout and Lahn, 2015). For water volumes measured at Kut, just north of the Al Hawizeh marshes, flow dropped from an average of 24.5 BCM per year over the 1953-1984 period to 13.9 BCM per year over the 1985-2005 period, a reduction of 60% (UNESCWA and BGR, 2013b).

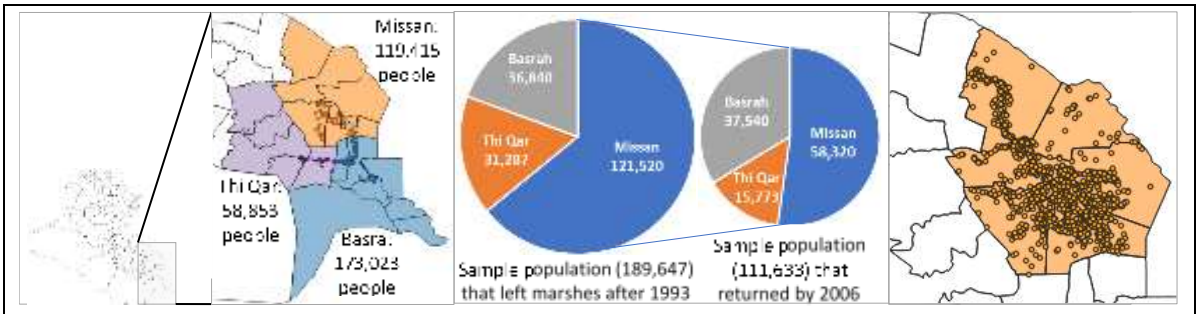


Figure 3: Left: Location and population of sampled villages in 2006 (Thi Qar University & UNEP, 2007); Middle: Breakdown of displaced and returning marsh dwellers (Thi Qar University & UNEP, 2007); Right: Location of all villages in Missan governorate as of June 2019 (UNOCHA, 2020)

7. In addition to these major upstream damming operations, systematic draining of the marshes for military and political reasons under the Saddam Hussein regime began at the conclusion of the first Gulf War. Large water retention structures were built to divert water away from the marshes in retribution for an uprising of the marsh population, further disrupting the hydrological balance of the region. During the 1990-2003 period, over 90% of the original marshlands were drained and subjected to systematic over-exploitation and a lack of coordinated management (UN, 2011). By 2003, the spatial extent of the Al Hawizeh marshes had been reduced by over 50%, with the Central and Al Hammar marshes having been completely drained (see Figure 2, top).

8. In the concerted retribution campaign against the Marsh Arabs, up to 200,000 were killed in the process of draining the marshes, or about 40% of the population of the region prior to the first Gulf War. The lives of the remaining marsh dwellers suffered mass displacement and egregious human rights violations, becoming Internally Displaced Persons (IDPs) subject to forced migration throughout Iraq and beyond. The combination of massive hydrological engineering projects, shelling and burning of marsh villages, the poisoning of fishing grounds and the assassination and abduction of local leaders, nearly totally destroyed the once-thriving habitat of the marshes and reduced the population of the marshes to an estimated low of 20,000 people living in a severely degraded environment amidst conditions of deepening poverty.

9. In recognition of the compelling need to reverse the environmental harm to marsh areas and to support the return of marsh IDPs, significant restoration efforts began in 2003 after the fall of the Saddam Hussein regime. This was considered essential not only for those IDPs who wished to return, but likewise for all Marsh Arabs in order to maintain vital community, social and cultural links. Just before the start of large-scale marsh reflooding and restoration efforts, it is estimated that a total of 40,000 people were living in the marshes (Thompson, 2003; SCECI, 2006).

10. The actual start of marsh restoration efforts began when returning and remaining marsh dwellers launched a grass roots campaign to demolish upstream water diversion/retention structures. These efforts were ad hoc and unplanned yet helped to return significant amounts of water to the region. Subsequently, restoration and management efforts have been implemented by and coordinated across multilateral organizations (e.g., UNEP, 2009) and bilateral agencies (e.g., USAID, 2004; CIMI, 2010). The results have been variable, with successful rehydration of some areas and restoration failure in other areas due to persistent high soil/water salinity and pollution.

11. By 2005, two years into restoration efforts, the marshes had been restored to about 41% of their pre-1973 area (see Figure 2, bottom). Since then, continuing restoration efforts have resulted in the increase of the area of the marshes by 2013 to about 65% of their pre-1973 size, with the greatest restoration results evident for the Al Hawizeh marshes (RoI, 2014). However, the Iraqi government's original goals to restore 75% of the 1973 Marsh extent is widely viewed as unrealistic, as there is not enough available water to meet restoration goals due to ongoing upstream damming activities and proposals (Chenoweth et al. 2011).

12. Efforts to rrestore the marshes were accompanied by a return of many marsh IDPs to the region. In 2006, a survey was undertaken to assess livelihood conditions for a sample of villages across marsh areas (Thi Qar University & UNEP, 2007). A total of 199 villages were selected in Missan, Thi Qar and Basrah governorates to document total population, migration levels, number of returnees, available infrastructure, as well as numerous livelihood conditions. By 2006, there were 346,291 people living in the sample villages (see Figure 3, left for the locations of these villages and corresponding governorate-level populations in the sample). In absolute terms, the Al Hawizeh marshes in the Missan governorate accounted for the largest number of IDPs who left after 1993 (121,520 individuals) of which about half had returned by 2006 (58,320 individuals); across the three governorates, about 60% or 111,633 of the sample population were returnees who had left just after 1993 (see Figure 3, middle). While no official current population count by village is available, by June of 2019 there were 1,293 inhabited villages in the Missan governorate within, bordering, and in the general vicinity of the Al Hawizeh marshes (see Figure 3, right).

13. As part of a 2019 national survey conducted by the Central statistical organization of the Ministry of Planning, nine (9) villages within the Al Hawizeh marshes were included in the sample. The survey results revealed that there are about 1,931 people per village, with roughly 224 households per village and an average household size of 8.6 persons.

14. Census demographic information at the Missan governorate level is summarized in Table 1, disaggregated by gender, for the year 2017. Most of the population (74%) is classified as urban dwellers, with females accounting for just over 50% of the total governorate population and about 51% of its rural population. Women account for only 38% of the skilled labor force in the region with most (60%) of the 2,733 female workers associated with the medical field in the form of nurses, doctors, pharmacists, etc. Women have a longer life expectancy at birth, 78 years, exceeding that of men by about 5 years. The maternal mortality rate per 100,000 live births in Missan is 42.5 which exceeds the national average of 31.0 but is below the value of 57 in 2017 for the Middle East and North Africa region and well below the value of 211 in 2017 for the world (UNICEF, 2020). However, women are more likely to be illiterate than men with only 32% of women over ten being literate compared to a figure of 68% of men. Additionally, girls represent only 13% of secondary school enrollees in the region (Missan governorate data).

15. Returnees to the marshes are taking up the same livelihoods as before, namely agriculture, livestock-raising, fishing, crafts, and commerce (Thi Qar and UNEP, 2007). These are practiced at a subsistence level, with animal husbandry constituting the main source of income with buffalo production and health having improved substantially, near doubling in population over the period 2006-2017 (Ayied and Reiss, 2019). Some key livelihood characteristics and indicators are summarized in the bullets below.

- *Agriculture.* Agricultural activities include the cultivation of rice, other cereals (i.e., wheat barley corn), vegetable, and fruit (i.e., dates and pears). Many marshes dwellers plant rice around the Tigris and Euphrates rivers, in addition to other crops like corn, wheat, barley, as well as watermelons, melons and cucumbers, tomatoes, and dates (Al-Tameemi, 2016). On a per capita basis, villages in the Missan governorate (Al Hawizeh marshes) are the most productive regarding cereal cultivation (i.e., 826 kg per year per capita across rice, wheat, barley, and corn) while villages in the Basrah governorate (Al Hammar marshes) are the most productive regarding vegetable and pears.

Table 1: Key demographic indicators for Missan governorate, 2017 (MoH&E, 2018a)

Indicator	Sub-indicator	Female	Male	Total	% of total
Population	**Total**	**543,940**	**539,997**	**1,083,937**	**100%**
	Urban	400,034	400,594	800,628	74%
	Rural	143,906	139,403	283,309	26%
	Older than 60 years old	26,848	23,191	50,039	5%
	15-60 years old	279,333	274,270	553,603	51%
	Younger than 15 years old	237,759	242,536	480,295	44%
	Adolescents (10-19 years old)	139,792	132,855	272,647	25%
	Younger than 5 years old	80,578	86,546	167,124	15%
	Child-bearing, 15-49 years old	254,973			24%
Skilled labor	**Total**	**2,733**	**4,476**	**7,209**	**100.0%**
	Administrative, finance, legal and service staff	1,080	1,384	2,464	34.2%
	Medical workers	1636	3021	4,657	64.6%
	Veterinarians	1	7	8	0.1%
	Statisticians	1	11	12	0.2%
	Engineers	11	7	18	0.2%
	Technicians	4	46	50	0.7%
Public Health	Total births	18,342	19,294	37,636	3.5%
	Life expectancy at birth (years)	78	73	75.4	
	Life expectancy at 60 (years)	21.6	19.5	20.5	
	Birth rate per 1,000 people in Missan	16.9	17.8	34.7	
	Death rate per 1,000 people in Missan	1.5	1.8	3.3	
	Neonatal mortality rate per 1,000 live births in Missan	7.9	12.2	10.1	
	Infant mortality rate per 1,000 live births in Missan	10.5	14.4	12.5	
	Under 5 mortality rate per 1,000 live births in Missan	14.9	18.9	17.0	
	Maternal mortality rate per 100,000 live births in Missan	42.5			

- *Livestock-raising.* Animal husbandry involves the raising of buffaloes, cows, and sheep. On a per capita basis, villages in the Al Hawizeh marshes in Missan have the highest number of cows and sheep, between 2 and 4 times as many as in other marsh villages. Buffaloes, valued for their adaptability to the marsh environment, provide marsh dwellers with milk and its derivatives, as well as meat (Al-Tameemi, 2016). Ownership levels are fairly consistent across the marshes, averaging about 0.6 buffalo per person.
- *Fishing.* While fishing has traditionally been an important contributor to the economic well-being of the marshes, commercial fishing has diminished considerably both in catch volume and diversity due to poor water quality (UNEP, 2009). Household engagement in fishing activities varies considerably across the marshes. In 2007, fishing was a common activity for 35%, 20%, 32% of the marsh households in the Thi Qar, Missan, and Basrah governorates, respectively (Thi Qar University & UNEP, 2007).
- *Crafts.* Traditionally, handicrafts made of reeds were a major part of marsh lifestyle and economy, commonly performed by women (Fawzi, et al., 2016; Al Tameemi, 2016; USAID, 2004). In Thi-Qar regions, reed products are common whereas in Missan, weaving wool carpets is evident (UNEP 2009). In the Hammar marshes, woven items such as mats, hand fans, serving platters, baskets for dates and other foods are typically made from reeds or date palm fronds (Fawzi, et al., 2016).

- *Commerce.* Local industries include boatbuilding from reeds (Al-Tameemi, 2016). Small canoes are the most important form of transportation in the marshes as they can be easily navigated through the reeds. They are made of wood and reeds and coated with bitumen on the outside. The boats are used for transporting people, and goods for selling.

- *Sandstorms.* The frequency of dust and sandstorms have also been increasing across Iraq. These are dry and dusty winds from the northwest with occasional gusts that can reach 108 km/hour at 300 meters above ground (Sissakian et al., 2013; Yu et al., 2015). They occur from April to early June and again from later September through November. This wind brings with it violent dust storms that may rise to heights of several thousand meters. While the maximum number of annual dust storms during 1951-1990 was about 24, within the coming years Iraq could witness 300 sand dust storms per year (Sissakian et al., 2013).

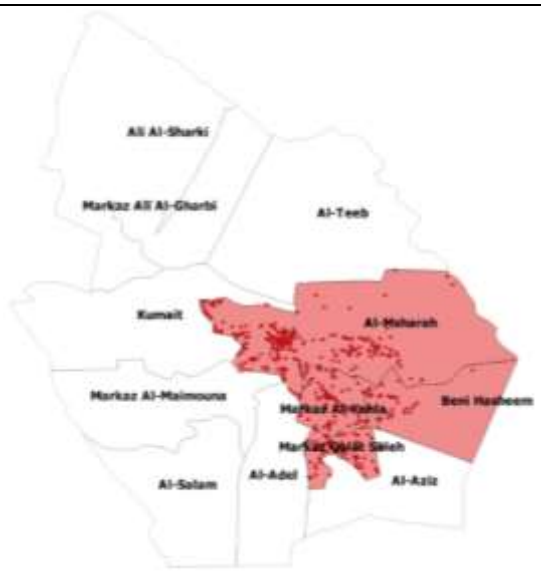
  Some of the main reasons behind the development of such storms are the climatic changes within the region, especially decreases in annual rainfall, as well as the drying of the marshes (Ministry of Foreign Affairs of the Netherlands, 2018).

- *Droughts.* Drought episodes are common throughout Iraq and have produced enormous economic, environmental, and social impacts. For marsh areas, the droughts that occurred during the growing seasons of 2007-2009, 2010-2011 and 2017-2018 had a devasting impact on local agriculture and livestock in the marshes forcing many households to leave their communities in search of better livelihoods elsewhere (Flintoff, 2009). Drought episodes are also increasing in the marshes in recent decades. In the Basrah region, the average number of drought episodes per decade increased from 22 during 1990-2000 to 35 during 2000-2010 (Agha and Sarlak, 2017). It is important to note that climate-induced displacement has produced some unexpected benefits in the form of increased awareness of new technologies and strategies that are used in cities and could be adapted and applied at the marsh village level. Efforts have been underway to deal with the increasing threat of droughts by the development of drought risk management plans in coordination with multilateral organizations (UNDP, 2012; UNESCO, 2014; MOST, 2015).

- *Biodiversity.* The capacity of the Al Hawizeh marsh to provide ecosystem services has been severely compromised by past destructive policies and which are now being exacerbated by climate change. The area represents some of the most critical wintering areas for migrating birds and is highly susceptible to increased soil salinization. Declining water flows into the marshes, together with increasing drought periods, have led to reduced permanent marshland and increasing water salinity, which in turn have led to the destruction of many flora and fauna species. Moreover, the reduction in waterflow has led to seawater intrusion, which is increasingly pushing inland – up to 45 miles upriver and as high as Basra at high tide. Water quality has therefore become a serious issue, having deteriorated substantially over the past decades due to changes in quality and flow of the Tigris and Euphrates rivers, rendering water largely unfit for human consumption, agriculture and other uses (UNEP, 2009). Fish stocks have been adversely affected by a doubling of salinity levels from intrusion of highly saline Arabian Gulf waters as well as increased algae levels due to the presence of phosphorus contaminants.

- *Public health.* Changing climatic conditions are aggravating the already poor water quality and wastewater management in the marshes resulting in increased risks of waterborne diseases - diarrhoea, cholera, typhoid, and different forms of hepatitis - relative to other parts of Iraq (UNESCO, 2014). Moreover, the dusty and dry conditions and the occurrence of wildfires irritate the airways and lungs resulting in aggravating chronic respiratory illnesses. According to the Ministry of Health, the highest number of diseases transmitted through contaminated water and food (Cholera, Typhoid, Dysentery, Hepatitis B) was registered between the years 2007 and 2010, during the most severe droughts that affected the country in the last decade.

Regarding COVID-19, an average of 70 people has been infected per day in Missan governorate, since the beginning of its outbreak in Iraq ,10 cases registered in village (Abu Khassaf) located near the Marshland. There are 79 health centres and 8 hospitals spread across Missan governorate, of the 12 villages located alongside the Marshland, only 4 of them contain a medical health center, while the rest of the villages are away from the health center by a distance of 5-15 km with serious health serious public health infrastructure challenges remain (MoH&E, 2018a).



Social
- 4,520 displaced families across 30 villages due to drought
- 903 farmers and breeders impacted by drought & water scarcity
- 4,068 farmers and breeders impacted by heavy rains & flooding

Agricultural
- 72 hectares lost of vegetable plots
- 4,880 kg of fruit destroyed
- Nearly 8,000 hectares of wheat plots destroyed
- Over 8,600 hectares of barley plots destroyed

Livestock
- 12,000 sheep lost
- Nearly 7,000 head lost (cows, sheep, goats and chickens)

Economic
- At least US$ 40,293 in lost income from sheep losses

Figure 4: Left: Sub-districts and villages in Missan governorate most affected by climate impacts over 2010-2018; Right: Types and levels of climatic impacts among the villages (source: Ministry of Health and Environment; personal communication)

16. There is a marked gender stratification associated with the above livelihood activities (MoH&E, 2018b). Women have a large role in family management and in some of the more onerous aspects of marshland daily life such as firewood gathering, water fetching, fodder collection, animal breeding, and the transport/sale of milk, fish, and chickens to/at local markets. In addition, all household chores such as cooking, and clothes washing are the sole province of women and girls. Men are mainly responsible for agricultural activities, hunting, house building, and some fodder collection.

17. In addition to the above impacts, heavy rainfall and flooding has increasingly plagued certain areas within the Al Hawizeh marshes over the period 2010-2018, leading to major social, livestock, agricultural, and economic losses, Among these communities, climate-induced disasters such as drought, flooding, and water scarcity have led to thousands of families being displaced, 4520 households distributed across two subdistricts were displaced due to drought on 2018 , thousands of farmers and animal breeders adversely affected, thousands of hectares of crops destroyed, the death of many types of livestock, and the loss of much-needed income (see Figure 4, right).

18. In the future, climate change is expected to lead to significant seasonal changes in temperature and precipitation in the marshes, as well as throughout Iraq. Minimum projected temperature increases in the marshes range between 0.4 and 1.6°C by 2030 under the A2 scenario, one of the increasingly likely higher greenhouse gas emission scenarios from the 4rth Assessment Report of the Intergovernmental Panel on Climate Change (see Figure 5, top). Regarding future precipitation in the marshes, rainfall is projected to increase slightly in the A2 scenario. The

average difference in seasonal rainfall between historical climate over the period 1975 – 2010 and the projected future climate over the period 2010–2040 period is in the range of a 0 to 20% increase under the A2 scenario for marsh areas. (see Figure 5, bottom).

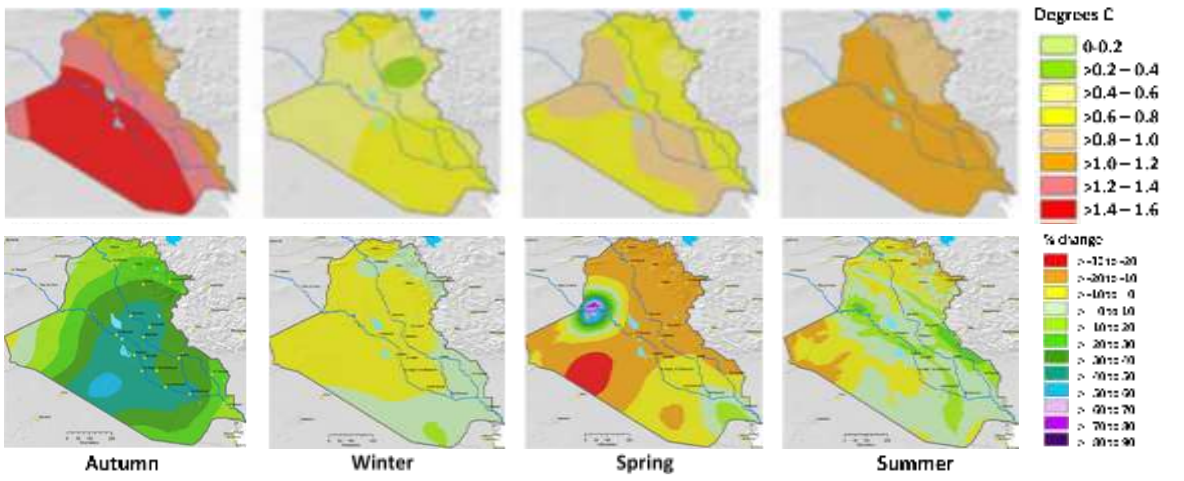


Figure 5: Projected temperature and precipitation change under the A2 Scenario. Top: Change in minimum seasonal temperature between current (1975-2010) and future (2010-2040) climate (De Pauw *et al.*, 2015); Bottom: Relative change in seasonal precipitation between current (1975-2010) and future (2010-2040) climate (De Pauw *et al.*, 2015)

19. It is important to note that there are areas in Iraq that are projected to experience future decreases in seasonal rainfall. These areas are mostly found in north eastern areas bordering Turkey and Iran as well as north western areas bordering Syria. Projected decreases of rainfall in these areas would adversely impact the marshes by reducing the future amount of upstream runoff into and subsequent water discharges into the marshes associated with the Tigris-Euphrates river basin.

20. In addition, projected surface runoff and discharge associated the Tigris and Euphrates river headwaters is projected to decrease. As part of the Regional Initiative for the Assessment of Climate Change Impacts on Water Resources and Socio-Economic Vulnerability in the Arab Region, a set of high-resolution regional climate models was used to downscale outputs from coarser general circulation models for the IPCC's Representative Concentration Pathway 8.5 (UNESCWA *et al.*, 2017). The results were then used as inputs to the Hydrological Predictions for the Environment (HYPE) model, a dynamic rainfall–runoff model which describes hydrological processes at catchment scale. The modelling results are illustrated in Figure 6 and show that runoff and discharge into both rivers is projected to significantly decline through 2100. The average change in discharge by the latter part of the 21st Century is between 20% and 30% for the Tigris and between 20% and 35% for the Euphrates.

21. Declining water flows into the marshlands, together with increasing frequency of recurrent drought periods, have led to reduced permanent marshland and increasing water salinity, which in turn have led to the destruction of many flora and fauna species. The loss of habitat for migratory birds caused some species to go locally extinct while commercial fisheries which used to account for around 60% of national fish consumption dropped precipitously (Al-Ansari et al., 2012). Moreover, the reduction in waterflow has led to seawater intrusion, which is increasingly pushing inland – up to 45 miles upriver and as high as Basra at high tide, additionally contributing to water and land degradation.

22. Water quality has therefore become a serious issue for the marshes. It has deteriorated substantially over the past decades due to changes in quality and flow of the Tigris and Euphrates rivers, rendering water largely unfit for human consumption, agriculture and other uses (UNEP, 2011). Along its route from eastern Turkey, alkalinity and pH deteriorates as it accumulates pollutants along the way including untreated domestic/industrial waste and pesticides. Salinity levels that have doubled owing to intrusion of highly saline Arabian Gulf waters as well as increased algae levels due to the presence of phosphorus contaminants have adversely impacted fish stocks. The effects of climate change across the headwater regions of the Tigris-Euphrates combined with ongoing upstream infrastructure development are expected to adversely impact water discharge levels of the Tigris and Euphrates on which marshland security and agriculture depends.

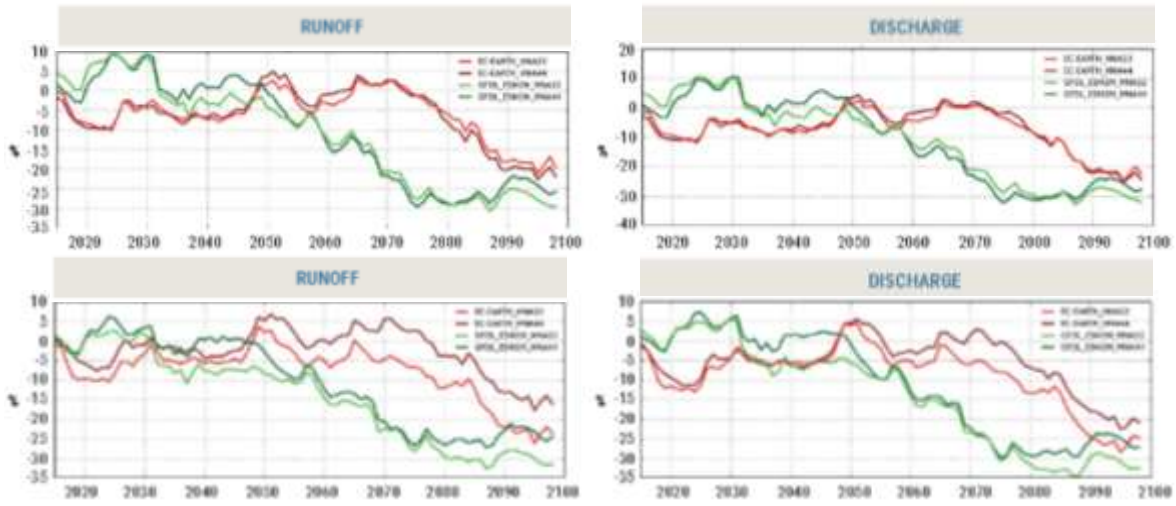


Figure 6: Top: Average change in runoff and discharge in Tigris River; Bottom: Average change in runoff and discharge in Euphrates River (UNESCWA, 2017)

23. In summary, climate change may well pose an existential threat to returning IDPs. Men will continue to be forced to leave the marsh homelands and their traditional occupations to seek employment in urban centers, leading to further fracturing of family and social networks. With rising temperatures and reduced upstream headwater flows, climate change will lead to numerous adverse impacts on returning communities including exacerbation of water pollution/scarcity, intensification of disaster shocks, and threats to public health. The combination of these factors layered onto the baseline socioeconomic fragility of returning IDPs will further undermine social cohesion in the marshlands as many communities can be expected to experience an intensification of societal disruptions, property losses, and livelihood uncertainty.

II. Strategy

24. UNDP Iraq seeks to implement a 3-year Climate Action for Human Security in Iraqi Marshlands project, jointly with the Iraqi Ministry of Health and Environment and the local government in Missan. The project strategy is a direct response to the current levels of severe environmental degradation in the Al Hawizeh marshes, amidst returning IDPs, as discussed above. The projected changes in marsh and upstream climate regimes may well reverse the pattern of return of IDPs to the marshes and induce a second wave of displacement over time due to the deterioration of marshland biodiversity and the ecosystem services on which the livelihoods of returning IDPs depend.

25. Based on consultations with the government, local communities, and civil society stakeholders on climate risks in the marshlands and priority activities to reduce displacement pressure, the overall objective of the project is to enhance climate security among returning IDPs to the Al Hawizeh marshes.

26. The project's implementation strategy is to support crisis-affected communities with a special focus on women's empowerment through a) deployment of innovative clean drinking water supply and wastewater treatment systems, and b) support for livelihood diversification and ecotourism activities consistent with and in coordination with government's natural resource management plans. Moreover, the project is designed to directly address the previously discussed interlinked development challenges as briefly outline in the bullets below:

- *Social cohesion:* An important element of the project design is to promote social cohesion in the marshlands. Marsh communities have experienced human rights violations, social network disruptions, property loss, social tension, and uncertainty. Upon their return from prolonged displacement, they now face harsh living conditions and other daunting social stressors that can contribute to increasing violence and mistrust. The proposed interventions are designed to enhance coping mechanisms among these affected communities, premised on the notion that the resulting improved well-being will positively enhance relations between and within communities. For example, by establishing women cooperatives for crafts production, marketing and saving schemes, the project will contribute to interdependence and social cohesion within and between communities.

- *Gender equality:* Conflict, internal displacement, deepening poverty and weakened state institutions have made Iraqi women and girls particularly vulnerable - to domestic violence, to barriers limiting participation in civil society, to a widening gender gap. Large inequalities still exist in the marshlands, with women systematically denied the same work rights as men and experiencing discrimination in public affairs. The failure to engage women in local decision making is simultaneously depriving the region of ideas based on community needs. Climatic shocks will also have a disproportionate effect on women and children. The project's design addresses gender equality issues through the development of a gender assessment and action plan that will outline measures to promote participation and economic empowerment by surveying women's perspectives to better understand household water supply and sanitation needs as well as to understand intra-household and community gender relations with a view to enhancing status of women in their communities. The gender action plan will also ensure that women are engaged in decision making and other aspects of project related activities.

- *Women's agency local (NGOs):* The project's design seeks to ensure women's agency and voice in all its components and outcomes. As some women in Missan Province have been active in NGOs working in the marshlands, their engagement will be integrated into the project's awareness-raising and capacity building activities that involve NGOs, including community-level organizing for advocacy on climate change adaptation in the marshes and seeking to account for climate change in future planning in the marshes. Community engagement and empowering women as agents of change will be integrated in all project outcomes

- *Stability:* Marshland communities have been severely damaged by an IDP crisis, environmental deterioration, and recovering national institutions. Even as the IDP return process is well underway in the marshlands, there remains a larger concern that the lack of infrastructure and social services will plague local communities for years to come and contribute to increasing social instability. The project's design directly addresses stability issues through providing sustainable water and sanitation infrastructure, promoting ecotourism, and reducing gender inequality.

27. The project is consistent with Iraq's National Development Plan (NDP) for 2018-2022 which emphasizes that Arab countries like Iraq will be at the heart of major climate challenges facing the world by 2050. The project is also consistent with Iraq's Intended Nationally Determined Contribution (INDC) under the Paris Agreement which calls for the recovery of marshlands and their communities. The design of the project is based on national and marsh-level consultations on climate risks in which a baseline situation was established: marsh communities will increasingly suffer from inadequate water supply and waste treatment infrastructure while living within a fragile and degraded environment unable to withstand and recover from climatic shocks, particularly sandstorms and drought. Absent effective adaptation measures, such conditions work against ongoing efforts to protect and restore marsh areas and will worsen over time.

28. The project will work to enhance social cohesion and stability towards SDG-16, enhance resilience and strengthen local climate action towards SDG-13, and improve the provision of clean drinking water and sanitation towards SDG-6. In addition, the project tackles SDG-5 on gender equality and SDG-1 on poverty which go hand-in-hand with climate security and resilience in the Marshlands. Given the close proximity to the national border with Iran, the pilot project will also align with and reflect the adequate consideration of the Convention on the Protection and Use of Transboundary Watercourses and International Lakes (Water Convention) adopted in Helsinki in 1992.

29. Specifically, the project takes a nexus approach to addressing climate change, deep poverty, water insecurity, and human displacement pressures by focusing on two complementary components that, when taken together, will result in strengthened Climate Action for Human Security. Detailed descriptions of each component are provided in the paragraphs that follow.

Component 1. Innovative solutions for clean water, wastewater treatment and climate security in Al Hawizeh marsh communities introduced

Overview

30. Component 1 will address two major objectives: clean water supply and effective wastewater treatment towards implementation of SDG-6. First, water scarcity has become the main reason for climate-related displacement, as per data on displacement collected by the Government and submitted to the Sendai Framework on Disaster Risk Reduction. This is relevant also in the context of implementing SDG-13 on climate action, SDG-16 on peace and the Paris Agreement. The demand for safe drinking water in the Al Hawizeh marshes is increasing due to the increasing population. Ensuring clean water availability will support the continued resettlement in marshland communities by those previously displaced. While social tensions between returning IDPs and those who stayed are unlikely given the societal homogeneity, any conflicts will be addressed in consultation with local governmental authorities, in coordination with local tribal laws and customs. At present, there is no clean water supply infrastructure and communities rely on either tanker water or marsh waters. Solar PV water purification systems offers a simple and cost-effective village-level approach to providing safe drinking water to households. Such systems have a small spatial footprint and can process and purify polluted marsh waters and store purified water within an adjacent water tank. They consist of an enclosed system that includes solar panels, a DC-AC inverter, a water purification system that uses a conventional membrane Process (i.e., reverse osmosis), a battery pack to store electricity produced by the solar panels, and the water tank (see Figure 7). The system can also purify harvested rainwater, boreholes, ponds, and rivers to fill the tank.



Figure 7: Diagram of a typical solar PV water purification system

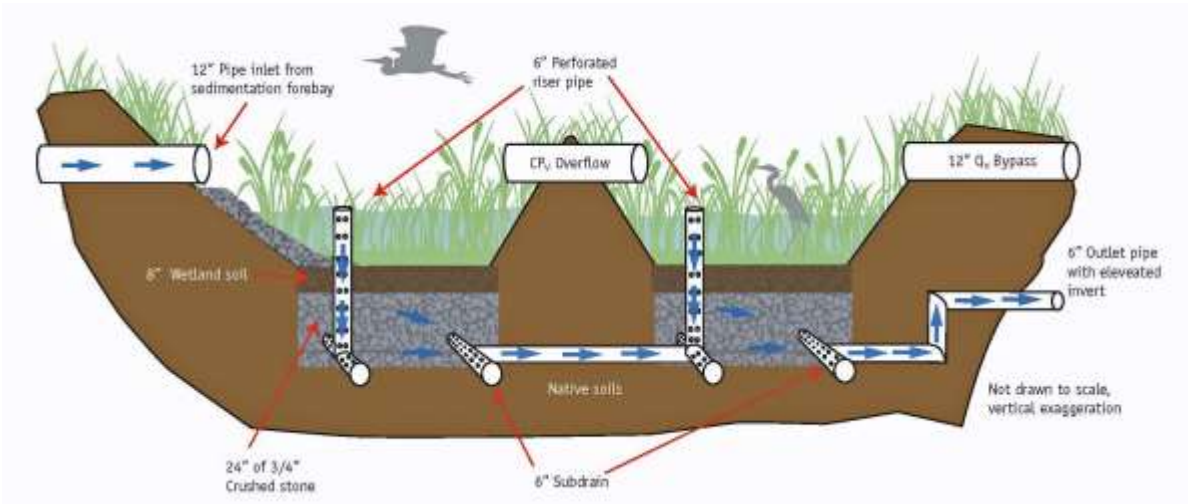


Figure 8: Diagram of a typically constructed wetland for wastewater treatment

31. Second, wastewater management exists at negligible to rudimentary levels in the Al Hawizeh marshes. Wastewater management through constructed wetlands is an economic and environmental strategy for a sustainable water treatment. The main advantages of such systems are their efficiency (combines biological, chemical and physical processes), the simple construction, the unlimited operating time and the low maintenance. There are three major components of a constructed wetland for wastewater treatment: excavated basins, crushed stone, an inlet pipe, and an outlet pipe (see Figure 8). While constructed wetlands are land-intensive, they offer an effective means of integrating wastewater treatment and resource enhancement, at a lower cost than conventional wastewater treatment alternatives. The water quality of the effluent from constructed wetlands is compatible with WHO drinking water standards.

Rationale

32. For a clean water supply, many water supply systems are commercially available at a cost between US$ 25 to 50 thousand, depending on the required daily volume of purified water production. For communities between 100-200 people, the specifications for a typical system consist of a 5kVA inverter, a solar PV array of 8.16 kWp, and battery storage equivalent to 50.88 kWh. The only input energy of the system is solar radiation at zero cost. Beyond the initial capital costs, the only other costs are associated with operations and maintenance activities, which are low, and the eventual replacement of the batteries at the end of their useful life.

33. For wastewater management through constructed wetlands, activities will be characterized by the following: a simple design as complex technological approaches will be difficult to maintain; and minimal maintenance systems that are designed to use natural energies, such as gravity flow. In addition, the system will be designed for climatic extremes, especially storms, floods, and droughts and consistent with the surrounding natural topography in order to mimic natural systems.

34. The design and implementation of effective cost sharing arrangements are essential to ensure that the project is more than a one-off intervention. Over the life of the project, operations and maintenance expenses will be borne out of the project budget. Over the long-term, operation of the facilities will be the responsibility of the local municipality while maintenance expenses will be borne by local community cooperative through a consensus strategy developed through a collective stakeholder process to ensure the financial sustainability of the installations. The community cooperatives, which will be comprised of at least 50% women members who will be empowered to propose suitable management agreements, will have the responsibility to design water use charges and water distribution arrangements compatible with community needs and constraints.

Outputs

In addition to the procurement and installation of off-grid solar PV system for purified drinking water,

Component 1 will include the following outputs:

A) Feasibility study. A total of 15 off-grid solar PV water purification systems and constructed wetlands will be installed. Of the 1,293 villages in Missan governorate, a total of 15 villages will be selected to introduce activities under Component 1. At the present time, the criteria for selecting target communities have not yet been identified, however, will include focus on returnees and communities that are prone to climate-induced displacement, among other factors.

Therefore, initial activities will focus on a) outreach to communities in the form of consultation workshops to explain the project, address questions, and seek community stakeholder engagement, while ensuring that the platform enables women to have a voice in the consultation

and decision-making process; and b) research into solar PV water purification systems commercially available in Iraq, as well as designs for constructed wetlands.

On this basis, a transparent village selection process will be developed using multi-criteria assessment of the ten indicators of the 2019 Multidimensional Poverty Index (MPI) using expert judgement and available gender-disaggregated data (as available), thereby targeting those communities most vulnerable to future displacement (UNDP, 2019). The technical feasibility of the technologies to satisfy drinking water requirements and water quality of the effluent from constructed wetlands will be determined by addressing site selection, willingness to pay (WTP), appropriate system sizing, water purification/treatment quality, suitable local vendors, associated cost-sharing arrangements in the mid- to long-term between government and local communities/entities, and other characteristics.

b) Gender assessment and action plan. Socioeconomic and climatic impacts are not gender neutral. Women and men, as well as boys and girls, use different measures to manage fragile social networks and adapt to changes in the natural environment. To understand the implications of the ongoing range of threats to social cohesion on the different roles and responsibilities of women - and to identify their respective needs to address them - the project will assess gender issues with a view toward developing gender-sensitive approaches and strengthening women's participation in decision-making. The gender assessment and action plan will provide an overview of the gender situation in the marsh villages where project interventions will occur and establish an action plan for integrating gender considerations in the preparation, implementation, monitoring and evaluation of all project activities (i.e., Components 1 and 2).

- Operations and maintenance (O&M). An effective O&M system enhances the likelihood that the installed systems will perform at or above their projected production rate and cost over time. Therefore, O&M procedures and system monitoring protocols will be developed to ensure long-term purification capacity and performance. At this time, solar PV O&M practices and approaches have not yet been standardized in Iraq so this activity will represent will be a significant contribution to decrease the perception of risk regarding similar future investments elsewhere in the marshes. In an official letter (see Annex 1), the Governor of Missan Province affirmed that water supply systems will be managed by local government water authorities in the long-term. Local village development committees will be established to raise funds for maintenance costs which are projected to be quite low. The project will also seek to include financing of the O&M costs in longer-term local development priorities and budgeting approaches, and develop a dedicated sustainability strategy to manage, maintain and finance sustainability of project results beyond the lifetime of the project.

- *A) Capacity building.* To ensure that there is a core set of villagers and governmental employees with the capacity to look after the installation, including O&M and monitoring protocols, a compact training programme will be developed directed at personnel at local government extension offices. The focus of the training will be twofold, namely on the attributes of solar PV energy water purification systems and how to teach this training content to marsh communities. The training will cover the essentials of solar PV technology, basics of PV site analysis, key tools and techniques for battery-based solar PV system maintenance, installation best practices, and essentials of water purification. Based on field surveys, the project will identify and include water supply and sanitation capacity building activities that specifically target women who will make up at least 50% of the trainees, as possible.

B) Awareness-raising campaigns. The aim of the campaigns is to increase awareness of the threat of climate change and highlight local climate resilience-building measures that also preserve and protect natural resources. Specifically, the awareness campaigns will spotlight ways of efficiently using water as well as conservation approaches. The overall participatory scope of the campaigns will be on civil

society with a particular focus on women and their unique role in marsh households in managing water use. The project will seek to employ youth organizations to support the awareness raising campaigns, as far as possible.

Component 2: Livelihood diversification and ecotourism to raise household income and protect ecosystem services, thereby enhancing resilience against future climate-related displacement among marsh communities, supported

Overview

35. The project will enhance livelihoods and income diversification and empower women by providing job opportunities through environmentally and socially responsible ecotourism in the Marshlands. Ecotourism in the Marshlands, if undertaken in a regulated and sound manner, has the potential to safeguard the integrity and diversity of underlying natural resources, provide education and recreation to tourists and national visitors, and increase and diversify income in rural areas, including for women-headed households, fishermen and women, and communities with the highest potential to be forcibly displaced due to climate-induced disasters.

Rationale

36. Ecotourism in the marshlands have the potential to build resilience to climate impacts in a fragile setting through developing greater social cohesion and promoting gender equality. Specifically, through job creation, income diversification, and the involvement of women, ecotourism activities have the potential to contribute to the establishment of enabling conditions conducive to long-term peace and stability. Moreover, the project's ecotourism activities build directly on the significant strides made in recent years to protect and conserve the marshes, thereby contributing to long-term ecosystem and livelihood resilience against climate shocks. In 2007, the Al Hawizeh marshes were designated as the Iraq's first Ramsar site, with an area of 1,377 km^2 (MoE, 2012). In 2013, with government endorsement of the Protected Areas Establishment and Management Bylaw Draft, four protected areas totaling 2,109 km^2 were established in the marshlands across the Thi Qar , Missan, and Basrah governorates, as illustrated in Figure 9 (left). In the same year, Iraq's first national park, the "Mesopotamia Marshland National Park" was established in a 1,416 km^2 expanse within portions of the Missan, Basrah, and Thi Qar governorates (see Figure 9, right). In addition, in 2016 the protected areas of the marshes became a United Nations Educational, Scientific and Cultural Organization (UNESCO) World heritage site.

37. Such Protected Areas (PAs) possess significant ecotourism potential. The flora of the marshes comprises nearly 60 plant species and represent one of the most important wintering areas for the many migrating water birds from Siberia and Northern Europe (Kubba and Salim, 2011). The current number of bird species in Iraq is 374, of which 311 are migratory birds (Birdlife International, 2019). Among this community, there are 17 globally threatened species of which 2 species – the Slender-billed Curlew (*Numenius tenuirostris*) and the Sociable Lapwing (*Vanellus gregarious*) are critically endangered. The marshes also offer wintering birds of prey such as rapters ample food, warmth, and protection.

38. Anecdotal evidence suggests that there is a growing local demand for marshland ecotourism. It is estimated that there were 10,000 tourists to the region in 2016, 12,000 in 2017 and 18,000 in 2018, almost all being Iraqi nationals. While such touristic activity represents a potential threat if unmanaged, it also represents an opportunity to achieve conservation goals and develop an important source of sustainable revenue for marsh dwellers. If managed well, nature-based tourism in the Al Hawizeh marshes can provide significant benefits, both for national/international conservation agencies and for local communities living in and around PAs. Nevertheless, ecotourism is at an early stage in Iraq as most initiatives are ad hoc with little to no strategic direction, with no investment in establishing sustainable management protocols, advertising, researching the market and technologies serving marshland ecotourism. To address this limitation, consultations will be undertaken with one or more ecotourism experts in order to propose a comprehensive strategy to promote ecotourism in the marshlands.

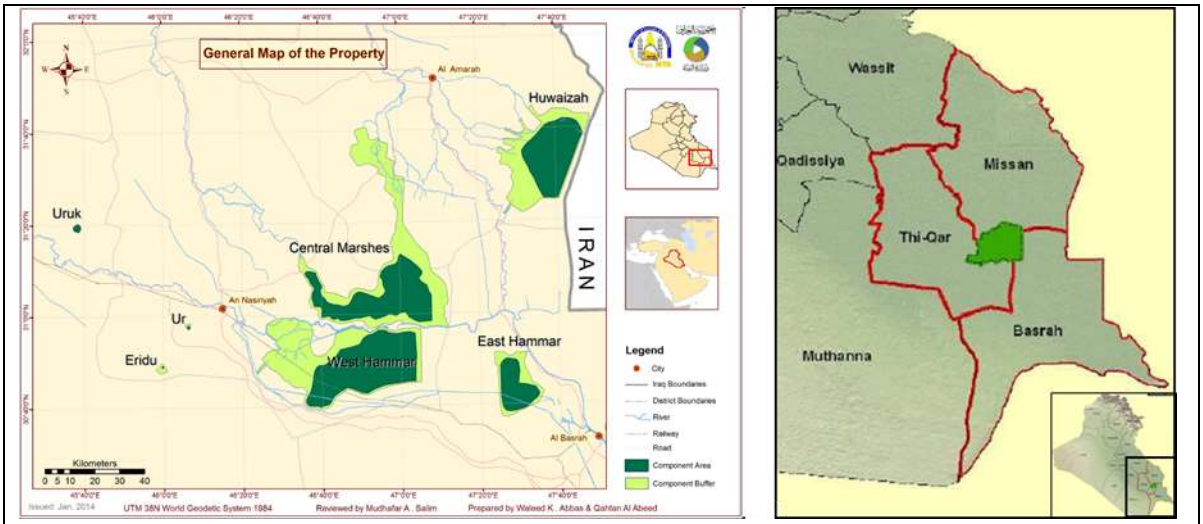


Figure 9: Left: Protected areas in the Iraqi marshes (RoI, 2014); right: Location of Mesopotamia Marshland National Park (RoI, 2014)

Outputs

39. Component 2 activities will be implemented in alignment with eight key principles, namely 1) minimize negative physical, social, behavioral, and psychological impacts; 2) build environmental and cultural awareness and respect; 3) provide positive experiences for both visitors and hosts; 4) provide direct financial benefits for conservation; 5) generate financial benefits for local people and the private sector; 6) deliver recreational value and experiences to visitors that help sensitize on climate impacts and linkages to local peace and stability; 7) design, construct and operate low-impact facilities; and 8) ensure that all activities are reflective of the gender action plan and include equal numbers of male and female beneficiaries. Based on these principles, the project will deliver three outputs as outlined in the bullets below.

- *Regulatory, environmental and investment guidance materials.* Such materials will be developed to foster a strategic approach to ecotourism activities in the marshes. These materials will represent an initial step in identifying the different types of attractions, development requirements, management restrictions and range of appropriate infrastructure investments. Particular focus will be on ecotourism as a conservation tool (e.g., stakeholders, protected areas, monitoring, marketing) and ecotourism management (e.g., range of

acceptable activities, lodging options, role of local communities, good practices in guest houses, promotion of ecotourism projects). On the basis of these guidance materials, two outputs will be generated. First, a detailed plan and environmental impact assessment will be developed for the type and location of ecotourism facilities to be introduced into the Al Hawizeh marshes, including but not limited to land-use zoning (e.g., dedicated areas for fishing), sustainable tourism minimum standards, and enforcement protocols for compliance with ecotourism guidelines. Second, a financial projection/analysis will be developed as a way to assess minimum touristic economic activity levels required to meet household income diversification benchmarks. The Ministry of Health and Environment will be the lead national institution involved in the development of the plan, in close coordination with local municipalities, invited local NGOs, and the project management unit. [2]

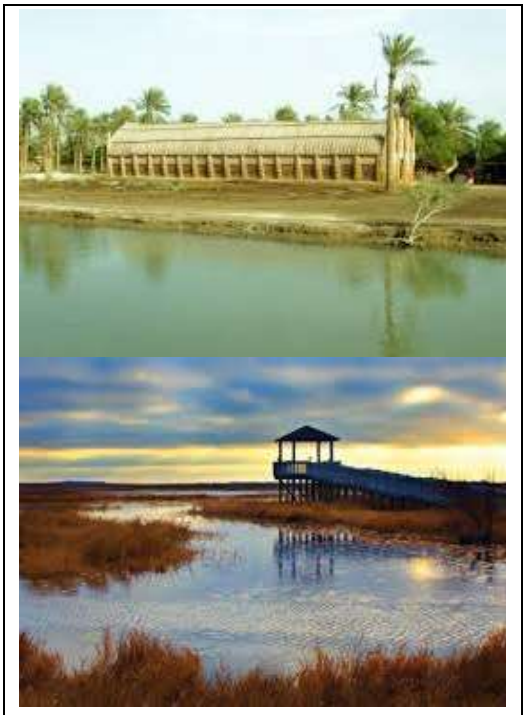


Figure 7: Illustrative examples of ecotourism facilities. Top: Welcome center; bottom: Wildlife observation towe

- *Construction of ecotourism facilities.* Second, a total of 6 ecotourism centers will be constructed in the Al Hawizeh marshes with the participation of local communities. While the precise type and location of these facilities will be established based on the detailed plan described above, they will be characterized by a minimal impact on natural resources and an emphasis on the protection of ecosystem services. The introduction of ecotourism will help diversify household income and provide additional livelihood opportunities for the local communities to increase their resilience and prevent displacement. The introduction of ecotourism will also focus on and help the most vulnerable groups in the marshlands (i.e., households whose living conditions are worse than existed before displacement), enhance gender equality (i.e., by focusing on women-headed households), strengthen social cohesion (i.e., by expanding and diversifying household income streams through a local, community-owned economic enterprise). Combined, ecotourism activities will contribute toward reducing fragility, preventing climate-induced displacement and enhancing human security.

  The core facility that will be constructed is a welcome center. It will be staffed by local stakeholders and offer welcome packets, group tours, and information of the physical and cultural heritage of the marshes. It will be constructed in an architectural style compatible with traditional structures (see Figure 10). The center will serve as a starting point for organizing guided tours on small eco-compatible boats through designated routes that do not interfere with the ecosystem. The center will also regulate touristic activities to limit impact on the ecosystem through designating areas for site-seeing. It will also provide training services to local women on handicraft manufacturing and market assessment for livelihood diversification. An off-grid solar PV system will be installed to provide electricity and water purification services. For each of the 6 ecotourism centers, a number of ancillary facilities

[2] While this effort will be led by the Ministry of Health and Environment, close coordination with the Ministry of Culture, Tourism and Antiquities will be sought.

will also be constructed, as briefly described below:

- ✓ *Guest house.* The guest house will be small roughly 10-room structure that is designed based on sustainable indigenous housing principles, respects the aesthetics of the marshes, retains local architectural styles and gives jobs to marsh dwellers as employees. It will be situated close to the welcome center and have its own off-grid solar PV electricity water purification system as well as basic sanitation infrastructure.
- ✓ *Market area.* A marketplace will be established in a suitably zoned area for marsh dwellers, especially women, to display and sell traditional crafts and products. The project's market-oriented activities will also seek to foster income diversification opportunities for women that are closely aligned and compatible with traditional practices such as agricultural and animal breeding activities, historically a source of income for women.
- ✓ *Dock for canoes.* The dock for canoes will be located in a suitably zoned area in close proximity to the welcome center and staffed by locals. It will offer guided tours as well as canoe rental options.
- ✓ *Wildlife observation towers*. Up to 5 wildlife observation towers will be installed in areas known for viewing migratory birds and other wildlife (see Figure 11, bottom). It will also visitors to look down and over the terrain rather than just across the horizon, where wildlife may be hidden in the ground cover.

- *Capacity building.* This will involve three major types of activities. First, awareness-raising & capacity building workshops among marsh communities regarding protecting biodiversity and ecosystem function in the context of ecotourism activities. Local NGOs will be recruited based on UNDP procurement regulations to be engaged in the design and implementation of up to 5 ecotourism training workshops and a marsh-wide sustainable ecotourism awareness campaign.

  Second, women-targeted training will be carried out on how to build on existing skills and resources to ensure women are able to carry out various activities beyond the project duration such as boat operations, fishing, tour guiding aimed at women and foreigners, or other culturally considerate activities that build on existing/learned skills such as high-quality crafts production. Third, capacity building activities will include the development of at least 3 locally driven, market-oriented innovative ideas that could promote ecotourism consistent with building sustainable livelihoods and promoting climate resilience. These training activities will also emphasize the development of new marketing skills among women and youth. Capacity building will also be provided to establish and/or strengthen women cooperatives for crafts and facilitate their access to markets, as well as support women cooperatives' saving schemes that enhances women empowerment and contributes to long-term sustainability of the project results beyond project completion.

III. Sustainability

40. The above project components and activities are closely aligned with the objective of the regional project "SDG Climate Facility: Climate Action for Human Security" (UNDP Project 00111749). Through its focus on supporting marsh communities to become resilient to growing risks from climate change and climatic disasters, the project is consistent with meeting a critical agenda item in the Arab region to address climate change as a threat multiplier to underlying poverty, human displacement, and water insecurity. The project's added value lies in its unique approach in addressing the nexus between climate action and human security through a set of integrated activities involving ecotourism, renewable energy solutions for water supply, and promoting livelihood diversification, especially among women. Upon completion, the project's outcome will include an increased awareness and understanding of the benefits of a nexus

approach to strengthen climate security and enhance resilience to climate-induced displacement pressures.

41. Project activities will follow a gender approach and contribute to minimizing climate risks to some of the most vulnerable communities in Iraq. These include IDPs, people living below the poverty line, farmers and herders exploiting marginal lands in and around the Marshlands, as well as fishing communities in the Marshlands, and women-headed households. By its focus on enhancing water security and introducing household income diversification opportunities, the project will reduce pressures associated with climate-induced displacement of households and communities and help build climate-resilient communities to better cope with and recover from climate disasters.

42. The project's sustainability is rooted in its approach to stakeholder engagement and participation. Engaging both marsh dwellers and staff within national agencies will promote ownership among local communities and extension authorities. The sustainability of the project is further enhanced through the efforts to develop institutional handbooks (i.e., regulatory/environmental guidance materials for ecotourism in marshes) and operational standards (i.e., water purification O&M protocols) that will contribute towards mainstreaming practices into institutional policy and strategic plans. Project sustainability will be further strengthened through the training and awareness-raising programmes that will be implemented for Components 1 and 2. Finally, the project is intended to be a way to pilot coordination mechanisms between the local government and federal authorities to address the development of marshland water supply codes/standards and ecotourism strategies under Component 1 that could be a basis for application elsewhere in Iraq.

43. The project's cost sharing approach addresses temporality issues as well as capital investment and O&M issues. Regarding temporality, during the 3-year project implementation period all capital and O&M costs for all interventions will be borne by the project. Upon completion of the project, all O&M costs will be borne by local government budget in the long-term (see Missan governor's letter in Annex 1). Any post-project capital costs such as replacement batteries for the solar PV water supply systems or new audiovisual equipment for ecotourism facilities will be borne by a combination of local communities/entities and local government as described in the feasibility study[3] under Component 1.

44. After closing of the project all equipment recruited for water supply system will be received officially by the authorized local governmental entities, and become governmental assets, this will ensure their working Permanence. Regarding ecotourism facilities, it will be owned by the beneficiaries from the local community, the project will encourage the local beneficiaries to elect a committee/association from them to look after the operation and maintenance of these facilities in cooperation with the local authorities.

45. Two types of sustainability provide the basis for the project's exit strategy. First, the project's internal sustainability is based on the fact that O&M for community-based solar PV water purification systems beyond the project implementation period will be fully borne by the government through governorate level extension offices. Capacity building activities and the development of technical O&M manuals will be instrumental in ensuring that these systems are sustained in the long term in all of the project target sites. The internal sustainability of the ecotourism interventions is premised in the income-generating nature of this component to support future investments to maintain facilities and expand services. The project's external sustainability is premised on the replication potential community-based solar PV water

[3] The feasibility study will include a sustainability strategy that will cover, among other things, financial sustainability planning.

purification systems. The 15 pilot installations have the means of scale-up as other communities become aware of the benefits of such systems or self-organize to invest in similar systems without additional resources being provided. The project will ensure that the do-no-harm principle of avoiding community exposure to additional risks is reflected by its emphasis on stepwise planning prior to interventions (i.e., technical feasibility study for Component 1, environmental impacts assessment for Component 2).

46. The project will contribute to specific Sustainable Development Goals (SGs) through its focus on climate action for human security. The project will produce co-benefits for the following SDGs:

- SDG-1 through income generation and livelihood diversification (Component 2);
- SDG-5 by empowering women and girls (Components 1 & 2);
- SDG-6 by providing clean drinking water (Component 1);
- SDG-7 by supporting clean energy solutions (Components 1 & 2);
- SDG-8 by strengthening economic growth of communities (Component 2);
- SDG-11 by supporting local cultural heritage in the Marshlands, which is UNESCO World Heritage (Component 2);
- SDG-12 by supporting sustainable eco-tourism (Component 2);
- SDG-13 by supporting clean energy for mitigation and water and community resilience as well as livelihood diversification for climate change adaptation (Components 1 & 2); and
- SDG-15 by supporting local communities to conserve the abundant biodiversity in the Marshlands (Component 2).
- SDG-16 by supporting social coherence and interdependence within and among communities (Component 2).

IV. Risks

Potential risks and mitigation measures are summarized in the table below. The overall risk rating for this project is considered to be between medium and high. As part of the Project's monitoring system, risks will be monitored and updated in ATLAS. on a quarterly basis and report on the status of risks which will be reported by Project manager as critical when both the impact and probability are high.

#	Description	Date Identified	Type	Impact (I) and Probability (P) 1 = none, 5= very high	Countermeasures / Management response	Responsibility	Submitted, updated by	Last Update
1	The security situation in Iraq remains unstable, including curfews due to public protests/riots, restricting travel, equipment transport, and work activities.	August 2020	Security	May prevent access to certain areas for implementation of projects I = 3; P = 3	Regular monitoring of overall security situation and abiding by UN Security Guidelines issued by UN Department of Safety and Security. Government remains operational despite the current political situation, with the UNDP Country Office able to effectively communicate with project counterparts on a daily basis. Missan and particularly the Project site is a secure place, with project endorsement by local tribal leaders	Deputy Resident Representative-Operations and Programme Security Team Project Manager	Project team	August 2020
2	The extreme heat and dust storms in the region may pose a risk for smooth installation of	August 2020	Environmental	May affect the efficiency of delivered solar power and, then affect the efficiency of water	UNDP has accumulated extensive project experience aligning technology choices and operating protocols with the requirements of Iraq's harsh environment, thereby	Project Manager	Project team	August 2020

	equipment, and its functioning leading to a potential loss of trust by target communities			supply system production I= 3; P = 2	effectively managing community expectations. Moreover, the proposed technical feasibility studies will identify and address maintenance requirements for long-term operation			
3	Difficulty for UNDP Project team to access the project sites to meet with local communities and authorities due to travel restrictions imposed due to the COVID-19 pandemic	Throughout implementation	Operational	May limit ability to achieve project results I = 3; P= 3	A National consultant will be recruited from the region to facilitate coordination and cooperation with local communities and authorities, thus limiting the impact of travel restrictions. Non-local members of the project team will take all appropriate precautions for traveling and undertaking local activities, including virtual meetings as needed	Project Manager	project team	August 2020
4	Some project activities (e.g., women's participation, ecotourism) may be incompatible with local customs and traditions	August 2020	Social	Negatively impact the achievement of quality results and targets for the relevant activities I = 3; P = 2	A local civil society organization will be recruited and tasked with designing approaches to securing community engagement in - and adoption of - all project activities, ensuring the principle of do no harm. This will involve developing an effective outreach/communication strategy, in a gender-responsive manner	Project Manager	Project Team	August 2020
5	Abrupt unobtainability of	August 2020	Operational	Lack of spare/tool parts will cause	Ensuring the integrity of the supply chain for post-project	Project Manager	Project Team	August 2020

	spare tools/parts for post-project operations and maintenance of clean water supply and wastewater treatment systems			**turning over of water supply and wastewater treatment.** **I = 3; P = 2**	**equipment (e.g., replacement batteries for the solar PV water supply systems or new audiovisual equipment for ecotourism facilities) will be addressed as part of the Operations and Management plan**			
6	**Corruption during procurement processes**	**August 2020**	**Financial**	**Negatively impacts the reputation of UNDP and the Project and results in a lack of trust in the Project.** **P=2** **I=4**	**All procurement actions will be managed in keeping with UNDP's Operational and Financial Rules and Regulations.** **The E-tendering platform will be used for the procurement process and obtaining required documentation.**	**Deputy Resident Representative** **Operations** **Head of Procurement unit**	**Project Team**	**August 2020**

V. Linkages with other activities

47. The project has strong linkages with a broad set of interventions in Iraq that aim to enhance climate resilience. These include: (i) UNDP's ongoing Climate Promise (USD 500,000) which will support the enhancement of Iraq's draft NDC, building national consensus and promoting commitment on ambitious low-emission and climate-resilient development; (ii) the project Building Resilience of Agriculture Sectors to Climate Change in Iraq (USD 9.9 million, IFAD, Adaptation Fund, 2018-2024); (iii) Iraq Crises Response and Resilience Programme (USD 138.5 million, UNDP, 2014-2021); and (iv) the Supporting Recovery and Stability Through Local Development in Iraq Project (USD 54.1 million, UNDP, 2019-2022).

48. There were several stages associated with the stakeholder consultative process for the development of this proposal, it was during drafting similar project on the south of Iraq Marshlands and Al Hewaizeh Marsh was part of it , First, key stakeholders across various stakeholder communities were identified and selected to take part in the consultative process. Second, a daylong project conceptualization workshop was held in Erbil in December 2019 to present and discuss potential adaptation project ideas and establish a priority set of project ideas for further consideration and refinement. Third, local consultations took place in January 2020 in Chibayiash village in the Central Marshes in Thiqar Province, the outcomes of these meetings contributed in developing and scope out potential sites of drinking water and wastewater treatment systems in this project

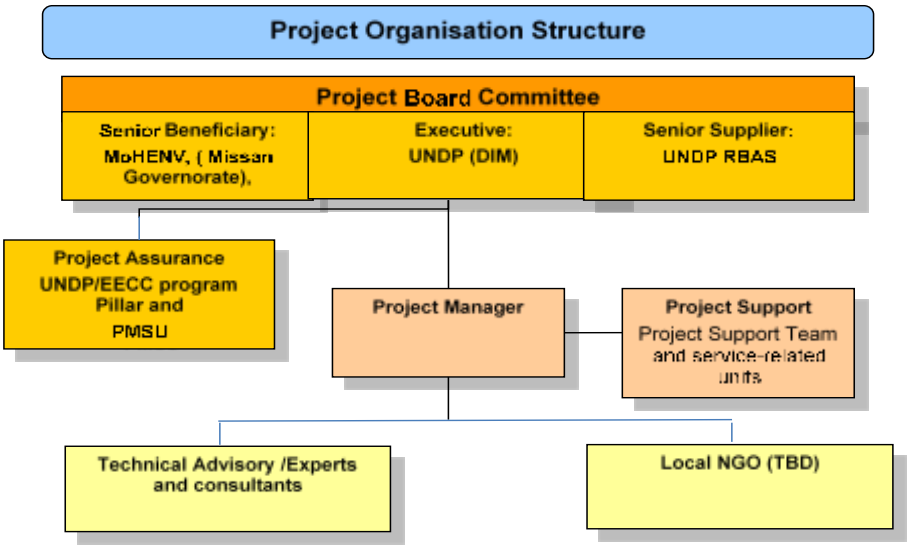
49. The project development also benefitted from consultations and input from key UN and government stakeholders, including the WFP Country Office, Sendai Framework Focal Point, Climate Change Focal Point, UN-Habitat Country Office, League of Arab States Sustainable Development Focal Point, and the Arab Water Council Focal Point. In addition, the Ministry of Health and Environment was represented by the focal points for the Marshland and the Sustainable Management of the Natural Environmental Resources Department; Climate Change Center /Adaptation section; and the International Relations department. The Poverty reduction strategy focal point from the Ministry of Planning was also a part of the consultative process.

50. The project also has strong linkages with national and sub-national sustainable development strategies and applicable laws. Specifically, the project's social cohesion, gender equality, peace/stability, and climate resilience framework is aligned with the following national initiatives:

- *New Eden master plan (2006).* The plan calls for supporting villages to provide a traditional community for returning refugees while providing modern services such as potable water, sanitation, health care, veterinary care, access to quick transportation, telephone, electricity, and even internet access. The "New Eden Village" implicitly advocates for an approach to sustaining livelihoods that has been adopted by the proposed project through its emphasis on supporting traditional lifestyle through the introduction of green technologies (e.g., decentralized water supply and sanitation systems) and benefiting from traditional environmental knowledge.

- *Draft Disaster risk management strategy in Iraq (2015).* Iraq's revised Disaster Risk Reduction strategy has been adopted by the National Security Council under the Prime Minister's Office. Prepared under the Committee of Order No. 101 by the Ministry of Science and Technology (MOST), the strategy calls for shifting away from a reactive posture towards disaster to a more proactive posture in which risk reduction is emphasized in line with the Hyogo Framework for Action document as well as the Arab Strategy for Disaster Risk Reduction which was adopted by the Arab Summit in Baghdad in 2012 (MOST, 2015). The Strategy's vision and goals are premised on four (4) priorities, namely understanding disaster risk; strengthening disaster risk governance; investing in disaster risk reduction; and enhancing disaster preparedness. The proposed project's emphasis on building resilience to climate change will mitigate current impacts and reduce future risks and is fully consistent with these governmental priorities.
- *Protected Areas Establishment and Management Bylaw Draft (2013).* This bylaw addressed guidelines and processes for establishing protected areas as well as their legal designation and management arrangements. The bylaw supports the maintenance of the conservation status of the protected areas by including provisions against land use change or harmful human activities (e.g., unsustainable agriculture, human settlement expansion). The bylaw also adopts a set of articles addressing the early warning systems and response plans. The proposed project is designed to be highly aligned with the bylaw's provisions on limiting the adverse impacts of community activities.
- *Ministry of Water Resources Law No.50 (2008).* This law addresses the water planning and investment context to ensure efficient use of water resources, as well as their sustainable development, enhancement, and usage. Within the law, priority is given to environmental protection aspects associated with the rehabilitation and restoration of the Marshlands as a water body of key national and international importance. The proposed project is designed to be specifically aligned with the bylaw's prohibitions on sustainable use of water resources.
- *Initial National Communication (INC) to the UNFCCC (2016).* The INC calls for several measures to adapt to climate change in the marshes, including removing threats to biodiversity and ecosystems (e.g., pollution discharges), reducing fragmentation of habitats, and developing strict water quality standards, among others. Project activities are designed to be compatible with these measures in the protection of water and biodiversity resources.
- *Ministry of Planning relief funds.* In August 2019, the Ministry of Planning and the UN signed today a Memorandum of Understanding (MoU) to establish the Iraq Reconstruction and Recovery Trust Fund in support of the Government's Recovery and Development Framework. Among other things, this represents a relief fund for rural villages across governorates to support the women and youth; the Missan governorate is covered by this fund.

VI. Project Management arrangement/ Monitoring and Evaluation



48. The project will be implemented through the Direct Implementation Modality (DIM) by which the UNDP-Iraq will be responsible for overall management and timely achievement of results. UNDP will work in in close coordination with the MoH&E and with Missan Governor's office, both of which will designate official focal points who will provide management support to UNDP for the implementation of project activities. The government focal Points will not be paid from project funds but will represent an in-kind contribution.

49. The project comes under the oversight of the UNDP Iraq's Environment, Energy and Climate Change Programme Pillar, which will provide Programme Quality Assurance. The project team will benefit from support services from the UNDP Country Office's Finance, Administration, Human Resources, Procurement Unit and the Programme Management Support Unit, as well as Country Office shared services such as IT, Communications and Security. The pilot project team, including a project officer for project management and a project officer for operation / finance, will perform daily project implementation.

    A project steering/board committee at the national level will be established to guide the implementation of the "Climate Action for Human Security in Iraq's Al Hawizeh Marshlands project." The project steering committee will provide strategic direction and oversight to ensure that the funded interventions are consistent with government and governorate priorities, project objectives and UNDP procedures. The project steering committee will be chaired by the Resident Representative- UNDP Iraq and will include the Focal Point from the MoH&E and a representative from the Missan governor's office as Senior Beneficiaries.

The project steering committee will be supported by a technical team consisting of key stakeholders from the following national Ministries: Water Resources; Agriculture; Tourism and Antiquities; and Planning. It will also be supported by key local government stakeholders in Missan as well as civil society stakeholders to ensure the technical, environmental, and social objectives of the project are achieved.

UNDP will prepare mid-annual and annual progress reports each calendar year, submitted to the project steering committee and to the regional umbrella project office of the "SDG Climate Facility Project: Climate Action for Human Security" at the latest 45 days after the end of reporting period. This will include updates on implementation progress, including capturing emerging issues and challenges and risks, that require the review and decisions by the Project Board. The Project Board will convene at least annually, although the Chair may decide to hold meetings at an ad-hoc basis as well. The Project Board is expected to play a facilitating role and serve as an agile mechanism providing timely guidance in rapidly changing contexts. Every effort will be made to achieve consensus on key issues. UNDP Iraq will also monitor and share information on the status of implementation with the SDG Climate Facility project team at the UNDP Regional Hub for Arab States at least every three months, including on results, potential challenges, and achieved milestones, to ensure implementation progress is in line with the regional project document and the agreed work plan in the pilot project proposal for Iraq.

50. The log frame that follows summarizes the project components, outputs, baseline, indicators and corresponding budgets to achieve the Outputs. The indicators are directly related to each output and can be measured objectively. Means of verification include workshop notes, document developed, structures/equipment installed, and awareness-raising campaign materials developed.

51. The project outputs as outlined in the log frame will be monitored and reported annually and evaluated twice during project implementation to ensure the project effectively achieves intended results. Project-level monitoring and evaluation (M&E) will be undertaken in compliance with UNDP's requirements for results-based management, and the UNDP Evaluation Policy. In addition to these mandatory M&E requirements, any other M&E activities deemed necessary to support project-level adaptive management will be agreed during project start-up activities and will be detailed in an Inception workshop Report. This will include the exact role of project target groups and other stakeholders in project M&E activities including national/regional institutions assigned to undertake project monitoring.

52. Upon project completion, an independent final evaluation will take place, the evaluation process will be planned and managed in keeping with UNDP's Independent Evaluation Guidelines and managed by the EECC Programme Pillar. The final evaluation process will begin at least three months before operational closure of the project, while the project team is still in place, yet ensuring the project is close enough to completion for the evaluation team to reach conclusions on key aspects such as project sustainability.

M&E work plan and budget

Type of M&E activity	Responsible Parties	Budget US$ *Excluding project team staff time*	Time frame
Inception Workshop and Report	Project Manager UNDP CO (including EECC Team)	Indicative cost: 2,000	Within first two months of project start up
Annual progress reports	Project manager and team UNDP CO (including EECC Team)	None	Annually
Progress Reports	Project manager and team	None	Bi-Annually
Final Evaluation	Project manager and team, UNDP CO (including EECC Team) UNDP EECC External Consultants (i.e. evaluation team)	Indicative cost: 10,680	At least three months before the end of project implementation
Project Terminal Report	Project manager and team UNDP CO local consultant	None	At least three months before the end of the project
Monitoring and Technical Quality Assurance Field Visits to project sites	Project Team UNDP CO (including EECC Team) UNDP PMSU (as appropriate) Government representatives	None	Regularly throughout the Project, as per a Annual Monitoring Plan
TOTAL indicative COST Excluding project team staff time and UNDP staff and travel expenses		US$ 12,680 (+/- 5% of total budget)	

Project Component/Outcome	Outputs	Links to Key Area	Baseline and Indicator	Timeline (in months)	Timeline (in years)			Cost (in US$)		
					Year 1	Year 2	Year 3	SDG Climate Facility Project Grant	UNDP Iraq Contribution	Total
Component 1: Innovative solar solutions for clean water, and climate security in Al Hawizeh marsh communities	**Output 1.1: Consultations and priority assessments completed** Conduct a Project Inception workshop to discuss stakeholder's rolls. Conduct a multi criteria capacity needs assessment for a transparent village selection process targeting the most vulnerable communities Conduct a site feasibility study for the project water supply interventions including field visits to the Al Hawizeh marshes; Develop financial sustainability plan within the feasibility study to ensure maintenance of project results beyond project completion; Conduct a Gender assessment and develop an action plan;	Key Area 1: ☒ Key Area 2: ☐	Baseline: Limited assessments completed for gender and water supply interventions in the Project location (August 2020) Indicator 1.1: # of assessments completed Target: Two assessment reports and one feasibility study developed.	9				62931	2000	64931
	Output 1.2: Off-grid solar PV system for drinking water and waste management operational Procure and install off-grid solar PV solar system for drinking water, and construct wetlands for waste management, Design Operations and Management and monitoring protocols.	Key Area 1: ☐ Key Area 2: ☒	Baseline: Absence of effective off-grid water supply infrastructure (August 2020) Indicator 1.2: Number of village-scale Solar PV water purification systems installed Target:15 systems installed, directly benefitting at least 1,500 people, half of whom are women	16				272076	7000	279076
	Output 1.3: Capacity building and awareness raising conducted Develop gender-responsive train-the-trainers programmes on solar PV energy systems; Develop gender-responsive awareness-raising campaigns with at least 50% participation by women on climate change and water conservation.	Key Area 1: ☒ Key Area 2: ☐	Baseline: Limited awareness of climate change impacts and water efficiency & conservation options measures (August 2020) Indicator 1.3 # of people benefitting from training workshops and a marsh-wide awareness campaign Target: At least 1000 people (50% women) - disaggregated by type of activity (training workshops and awareness campaign)	6				99692	10000	109692

Component 2: livelihood diversification and ecotourism to raise household income and protect ecosystem services, thereby enhancing resilience against future climate-related displacement among marsh communities, supported	**Output 2.1: Regulatory frameworks and guidance materials developed for livelihood diversification through promoting eco-tourism** Develop regulatory, environmental, and investment guidance materials for ecotourism in marshes in cooperation with relevant line ministries, including detailed plan and Environmental Impact Assessment for site location, and financial projection/analysis to assess investment requirements to promote household income diversification	Key Area 1: ☒ Key Area 2: ☐	Baseline: Limited national guidance materials for ecosystem activities in marshes (August 2020) Indicator 2.1: # of national guidance drafted/, developed or revised Target: At least two national guidelines (one developed) and one updated)	8				41538	15000	56538
	Output 2.2: Ecotourism facilities constructed Construction of ecotourism facilities in the Al Hawizeh marshes with the participation of local communities	Key Area 1: ☐ Key Area 2: ☒	Baseline: 0 ecotourism facilities in marshes (August 2020) Indicator 2.2: # of ecotourism facilities constructed Target: 6 appropriate tourism facilities compatible with local heritage, values and character constructed, directly benefitting at least 128 households who will experience reduced displacement pressures	16				202500	0	212500
	Output 2.3 Capacity of marsh communities developed Conduct capacity building initiatives among marsh communities regarding protecting biodiversity and ecosystem function in the context of ecotourism activities in addition to strengthen women capacities on providing job opportunities through environmentally and socially responsible ecotourism in the Marshlands	Key Area 1: ☒ Key Area 2: ☐	Baseline: 0 trainings on ecotourism workshops conducted (August 2020) Indicator 2.3.a: # of ecotourism training workshops conducted Target: 5 training workshops (at least 30% women) and At least 3 locally driven, market-oriented innovative ideas to promote ecotourism consistent with promoting climate resilience and protecting ecosystem services	6				152026	17000	169026
			Independent Final evaluation					10680	0	10680
			Subtotal					841443	51000	892443
			Project Management Cost					80557	99000	179557
			General Management Support Cost (GMS 8%)					68296	0	68296
			Total Project Cost (= Subtotal + Project management Cost + GMS 8%)					**922,000**	**150000**	**1072000**

Annex I: Official letter from Missan Governor's office



بسم الله الرحمن الرحيم

Republic of Iraq
Province Maysan
Governor's office

جمهورية العراق
محافظة ميسان
مكتب المحافظ

العدد : ٣١٠
التاريخ : ٤/١١/٢٠٢٠

NO:
Date:

الى / منظمة الامم المتحدة (UNDP)

م/ تقرير

السلام عليكم ورحمة الله وبركاته

رسالتكم الالكترونية الواردة الينا بتاريخ ٢٠٢٠/٣/٣١ المتضمنة تنفيذ مشروع اهوار ميسان ...

نود ان نبين لكم الاتي :

١- يجب ان يكون المشروع على مرحلتين :

- المرحلة الاولى : وهي اثبات وجود المشروع وقراءة المعطيات للمرحلة الثانية ويكون ذلك بانشاء محطة ماء صالح للشرب والاستخدام البشري والمياه الصالحة للمواشي ومن مصادر متعددة وتعمل المحطة على الطاقة الشمسية الواطيب توفرها وتسلم الى مديرية ماء ميسان بعد تدريب الكوادر عليها وتوفير المواد الاحتياطية وتغذي المؤسسات الموجودة في المنطقة ، وتشمل المرحلة كذلك تنفيذ مرسى الزوارق ومكان استراحة بمثابة مقهى او مطعم ومتحف بسنعة وقوف سيارات وتزويد اصحاب الزوارق بالخبرة تعمل على محركات (D.C) والبطاريات ومكان مخصص لشحنها.
- المرحلة الثانية : يجب ان تسبق هذه المرحلة دراسة جدوى حقيقية وتكون لمدة لا تقل عن ستة اشهر بعد انجاز المرحلة الاولى لغرض معرفة الالية التي يجب ان تنفذ بهذه المرحلة والتي تهتم بتوفير الجانب المعيشي الذي يساعد على استيطان اهالي المنطقة تشمل محلات بيع المنتجات الخاصة بالاهوار وتسويق الاسماك ومشتقات الحليب ويكون الهدف من الدراسة معرفة الكيفية التي يتم بها انجاح المشروع.

٢- ان تقام ورشة عمل في المنطقة للاطلاع وبشكل مباشر على طبيعة المنطقة والتداول مع اصحاب الشان من المختصين والمصممين للاستيضاح عن نمط المعيشة والاجواء والسبل الكفيلة لاختيار المواد المستخدمة في الانشاءات التي تتلائم وطبيعة المنطقة والمجتمع.

للتفضل بالاطلاع ... مع التقدير

علي دواي لازم
محافظ ميسان
٤/١١/٢٠٢٠

صورة عنه الى :

- مستشار المحافظ للشؤون الفنية / مطالعتكم في ٢٠٢٠/١/٣ للعلم مع التقدير
- الاضبارة العامة

حسين

Web site:www.maysan.gov.iq
Email:governor_office2006@yahoo.com

Annex II: Organizations consulted during proposal development

- **Yousif Muayad Yousif, Head of International Relations Department, Ministry of Environment and Health**
- **Mohammed Fadhil, Department of Marshlands, Ecosystems and Natural Resources, Ministry of Environment and Health**
- **Muayad Salim, Advisor to the Governor, Missan Governorate**
- **Samir Abood, Head of Environment Department in Missan, Ministry of Environment and Health**
- **Najla Ali, Director of the Social Fund for Development, focal point for League of Arab States, Ministry of Planning**
- **Hadi Hamdi, Deputy Director of Climate Change Center, Ministry of Health and Environment**
- **Alaa Saeed, focal point for Arab Water Council, Ministry of Water Resources**
- **Sahar Hussain Jasim, member of the national committee for Sendai Framework, IPCC focal point and Head of Adaptation Section, Ministry of Environment and Health**
- **Anna Soave, UN-Habitat Iraq**
- **Meenu Hada and Giorgi Dolidze, WFP Iraq**
- **Christoffer Burnett-Cargill, Embassy of Sweden, Baghdad**
- **William Evans, British Embassy Baghdad**

Annex III

List of references

Agha, O., and Şarlak, N., 2017. Analysis of meteorological drought in Iraq using the reconnaissance drought index (RDI). International Journal of Advanced Research 5(3):473-479.

Al-Tameemi, R., 2016. Reading the cultural specificities of the Iraqi marsh Arabs from their landscape, Landscape Research Record No. 5, Council of Educators in Landscape Architecture, Salt Lake City, Utah.

Buřič M., Bláhovec J., and Kouřil J. 2015. Back to the Roots: The Integration of a Constructed Wetland into a Recirculating Hatchery - A Case Study. PLoS ONE 10(4): e0123577. doi:10.1371/journal.pone.0123577

Canada-Iraq Marshlands Initiative (CIMI), 2010. Managing For Change: the Present and Future State of the Marshes of Southern Iraq.

De Pauw, E., Saba, M. and Ali, S. 2015. Mapping climate change in Iraq and Jordan, International Center for Agricultural Research in the Dry Areas.

Earth Link and Advanced Resources Development (ELARD), 2013. Preparation of drought vulnerability assessment study to develop Iraq national framework for integrated drought risk management (DRM) - Vulnerability assessment report, Draft report.

Fawzi, N. A.-M., K. P. Goodwin, B. A. Mahdi, and M. L. Stevens. 2016. Effects of Mesopotamian Marsh (Iraq) desiccation on the cultural knowledge and livelihood of Marsh Arab women. Ecosystem Health and Sustainability. 2(3):e01207. doi: 10.1002/ehs2.1207

Flintoff, C., 2009. Iraq's Marshlands Face A Second Death By Drought, news story for national public radio, available at https://www.npr.org/templates/story/story.php?storyId=101431344

Fund for Peace, 2019. Fragile States Index – Annual Report 2019. Available at https://fundforpeace.org/wp-content/uploads/2019/04/9511904-fragilestatesindex.pdf

Hashim, B., Maitham Abdullah Sultan, Mazin Najem Attyia, Ali A. Al Maliki, and Nadhir Al-Ansari, 2019. Change Detection and Impact of Climate Changes to Iraqi Southern Marshes Using Landsat 2 MSS, Landsat 8 OLI and Sentinel 2 MSI Data and GIS Applications, Applied Sciences, 14 pages.

Marjanizadeh, S., Qureshi, A., Turral, H., and Talebzadeh, P., 2009. From Mesopotamia to the third millennium: the historical trajectory of water development and use in the Karkheh River Basin, Iran. Colombo, Sri Lanka: International Water Management Institute. 51p. (IWMI Working Paper 135) doi:10.3910/2010.206.

Ministry of Foreign Affairs of the Netherlands, 2018. Climate change profile – Iraq.

Ministry of Health and Environment, 2018a (MoH&E). Annual statistical report, Republic of Iraq.

Ministry of Health and Environment, 2018b (MoH&E). Socio-economic status of Arab Marshlands in south of Iraq.

Ministry of Science and Technology, 2015. Draft National Disaster Management Strategy in Iraq (Revised), Prepared by the Committee of Order No. 101 of 2010, November.

RoI, 2006. New Eden master plan for integrated water resources management in the marshlands area, Volume 0, Main report, Ministries of Environment; Water resources; and Municipalities and Public Works

RoI, 2014. The management plans of the Ahwar property, Ministry of Environment.

RoI, 2016. Iraq's Initial National Communication to the UNFCCC, Ministry of Health and Environment.

Shamout, M and Lahn. G., 2015. The Euphrates in Crisis Channels of Cooperation for a Threatened River, A Chatham House research paper, Energy, Environment and Resources.

Sissakian, V., Al-Ansari, N., and Knutsson, S., 2013. Sand and dust storm events in Iraq, Natural Science, Vol.5, No.10, 1084-1094.

Special Committee on Environmental Cooperation for Iraq (SCECI), 2006. Report of the Special Committee on Environmental Cooperation for Iraq.

Thi Qar University and UNEP, 2007. Demographic, social and economic conditions of marshlands in south of Iraq, UNOPS Project No. 5119.

Thompson, L., 2003. Forgotten people: The Marsh Arabs of Iraq, Report from Refugees International.

UNDP, 2019. The 2019 Global Multidimensional Poverty Index (MPI), available at http://hdr.undp.org/en/2018-MPI

UNDP., 2012 Development of National Framework for Integrated Drought Risk Management (DRM) in Iraq, project document.

UNEP, 2009. Support for environmental management of the Iraqi Marshlands: 2004–2009. UNEP, Nairobi, Kenya.

UNESCWA and BGR, 2013b. Chapter 3: Tigris river basin, in Inventory of Shared Water Resources in Western Asia, Beirut.

United Nations Economic and Social Commission for Western Asia (UNESCWA) et al. 2017. Arab Climate Change Assessment Report – Main Report. Beirut, E/ESCWA/SDPD/2017/RICCAR/Report.

United Nations Economic and Social Commission for Western Asia (UNESCWA) and Bundesanstalt für Geowissenschaften und Rohstoffe (BGR), 2013a. Chapter 1: Euphrates river basin, in Inventory of Shared Water Resources in Western Asia, Beirut.

United Nations Educational, Scientific, and Cultural Organization (UNESCO), 2014. Integrated drought risk management – DRM: National framework for Iraq, Iraq office.

United Nations International Children's Fund (UNICEF), 2020. Maternal mortality statistics available at https://data.unicef.org/topic/maternal-health/maternal-mortality/#:~:text=Maternal%20mortality%20refers%20to%20deaths,to%20UN%20inter%2Dagency%20estimates

United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA), 2020. Dataset entitled: "irq_pplp_ocha_20140722.dbf" available at https://data.world/ocha-iraq/0c92944d-a6dd-4d08-a101-74ba1c2e3429

United States Agency for International Development (USAID), 2003. Strategies for assisting the marsh dwellers and restoring the marshlands in southern Iraq: Interim status report, Bureau for Asia and the Near East

USAID, 2004. The Iraq Marshlands Restoration Program, April.